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Summative Course Assessment

Graded Assignment: Course Summative Assessment Submitted

Your grade: 93.33%

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Instructions

1. Which statistical learning method is best for predicting a continuous outcome variable? 1 / 1 point

- ☐ Dimensionality Reduction
☐ Clustering
☒ Regression
☐ Classification

Assignment details
Due Dec 22, 11:59 PM ET 0 left 1 attempt every 8 hour Try again
Submitted Nov 17, 12:06 PM ET

2. What effect does multicollinearity have on the estimates produced by a linear regression model? 1 / 1 point

- ☐ It increases the overall fit of the model as measured by R-squared.
☒ It can make the coefficient estimates unstable. We keep your highest score.
☐ It reduces the computational time required to fit the model.
☐ It leads to a decrease in the standard error of the regression coefficients.

93.33%
Correct
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3. What does the slope in a simple linear regression model represent? 1 / 1 point

- ☐ The intercept of the regression line with the y-axis.
☐ The correlation coefficient between the dependent and independent variables.
☒ The rate at which the dependent variable changes for each unit change in the independent variable.
☐ The maximum value of the dependent variable.

Correct

4. In the context of smoothing splines, what is 'overfitting'? 1 / 1 point

- ☐ A situation where the spline is not smooth enough.
☐ Using a spline with too few knots.
☐ Fitting a spline to a dataset with insufficient data points.
☒ Fitting the spline too closely to the data, capturing noise rather than the underlying trend.

Correct
Correct! Overfitting occurs when the model captures noise instead of the true pattern.

5. What happens in the assignment step of the k-means algorithm? 1 / 1 point

- ☒ Each data point is assigned to the nearest cluster centroid.
☐ The value of 'k' is determined.
☐ The centroids are moved to the center of their clusters.
☐ The algorithm calculates the total within-cluster variation.

Correct
Correct! In the assignment step, data points are assigned to the cluster with the closest centroid.

6. In the context of subset selection, what is a potential disadvantage of using forward stepwise selection over best subset selection? 1 / 1 point

- ☐ It is more computationally intensive than best subset selection.
☐ It requires that the number of predictors be less than the number of observations.
☐ It evaluates too many models, leading to a high risk of overfitting.
☒ It might not include predictors that would be identified in a more exhaustive search.

Correct

7. What is a bootstrap interval? 1 / 1 point

- ☒ An interval estimate, such as a confidence interval, constructed using the bootstrap method.
☐ A fixed interval that determines how many data points are included in each bootstrap sample.
☐ The minimum and maximum values of the bootstrap samples.
☐ The interval between iterations in the bootstrapping process.

Correct
Correct! Bootstrap intervals are interval estimates derived from resampled data.

8. What is a common technique to assess if linear regression adequately models the dependent variable? 1 / 1 point

- ☒ Checking if the assumptions of linear regression are satisfied.
☐ Increasing the regularization parameter until the model's R-squared value peaks.
☐ Using only categorical predictors to simplify the interpretation of coefficients.
☐ Adding more predictors to the model until all p-values are significant.

Correct

9. In the context of machine learning models, how does the use of a kernel function typically impact model performance? 1 / 1 point

- ☒ It allows models to capture more complex patterns by transforming data into a higher-dimensional space.
☐ It directly reduces the training time of the model.
☐ It reduces the need for feature engineering.
☐ It simplifies the model to prevent overfitting.

Correct
Correct! Kernel functions enable models to handle more complex data patterns by mapping data into a higher-dimensional space.

10. Why should one study statistical learning? 1 / 1 point

- ☒ To build predictive models.
☐ To understand data visualization.
☐ To create large datasets.
☐ To become a statistician.

Correct

11. Why might a data scientist apply a square root transformation to a predictor in a linear regression model? 1 / 1 point

- ☐ To ensure that all regression coefficients will be squared.
☐ To increase the linearity of the relationship between predictor and response.
☐ To allow the regression model to fit polynomial relationships.
☒ To reduce the effect of skewness and extreme values in the predictor.

Correct

12. In the context of piecewise polynomial regression, what is a 'knot'? 1 / 1 point

- ☒ A point where the polynomial pieces connect or join together.
☐ A point of maximum curvature in the polynomial model.
☐ The highest degree of the polynomial used.
☐ A parameter that controls the overall fit of the model.

Correct
Correct! Knots are the points that separate different polynomial segments in piecewise regression.

13. In kernel smoothing, what is the effect of using a larger bandwidth? 1 / 1 point

- ☒ It results in a smoother estimate but may oversmooth important features in the data.
☐ It creates a more complex model with more peaks and valleys.
☐ It increases the sensitivity of the smoother to outliers.
☐ It reduces the computational complexity of the smoothing process.

Correct
Correct! Larger bandwidths in kernel smoothing produce smoother curves but can oversmooth.

14. Which of the following is a disadvantage of using tree models? 1 / 1 point

- ☐ Tree models cannot handle categorical variables.
☒ They can easily overfit the training data if not properly pruned or regularized.
☐ They are less interpretable compared to linear regression models.
☐ Tree models cannot be used for regression tasks.

Correct
Correct! Tree models can overfit if they grow too complex without constraints.

15. Which of the following is a potential issue when using maximum likelihood estimation for parameters of a statistical model? 0 / 1 point

- ☒ Maximum likelihood estimation always overfits the data.
☐ The likelihood function may have multiple maxima.
☐ The likelihood function always has a unique maximum.
☐ The likelihood function is always a convex function.

Incorrect
This is not necessarily true. Overfitting depends on the complexity of the model and the amount of data, not solely on the estimation method.